

SERAMAN - AI VIDEO PRODUCTION SYSTEM

System Handoff Documentation

Everything you need to run your automated video pipeline.

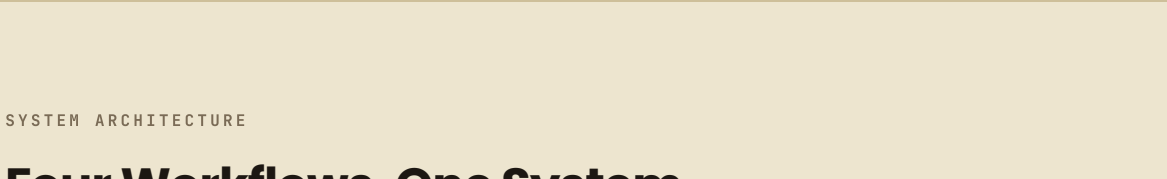
COMPLETE DELIVERY

DELIVERED	VERSION	PLATFORM	BUILT BY
June 2026	1.0	n8n + Kie AI + Creatomate	Emmanuel Adekoya

WHAT YOU NOW HAVE

One Form. One Video. Four Platforms.

Submit a product once. The system writes the Italian script, generates all video footage using Google's Veo 3.1 model, assembles the final video with captions, and publishes to Instagram Reels, TikTok, YouTube Shorts, and Facebook — automatically, in under 15 minutes.



No video editing software. No copywriting. No briefing calls. Every product in your catalog can now have a professional video within minutes of form submission.

HOW TO USE IT

Submitting a Product

The only thing you do manually is fill in the product form. Everything after that runs without you.

- Open your product form**
This is your permanent submission link. Bookmark it. Every new product starts here.
`https://tally.so/r/objx5vx`
- Fill in the product details**
Product name, Italian description (paste directly from your existing product copy), and the product image URL from your website or Tally storage.
- Submit**
The pipeline triggers instantly. The form validates your input — if anything is missing, you receive an email and the pipeline stops cleanly before spending any API credits.
- Wait for your delivery notification**
Within 10 to 15 minutes you receive an email confirming the video is live, with direct links to each published post on every platform.

SYSTEM ARCHITECTURE

Four Workflows, One System

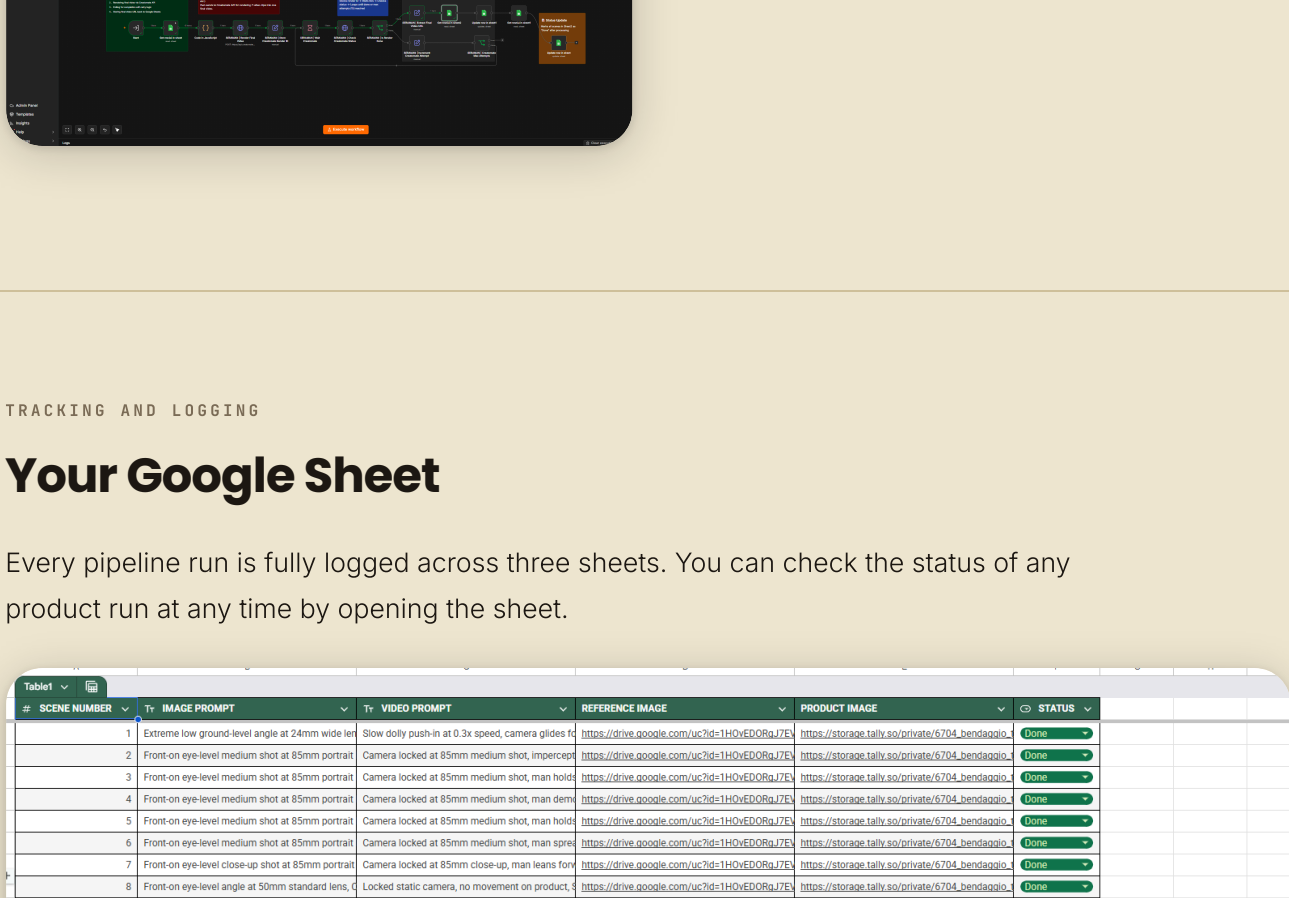
The pipeline is split across four modular n8n workflows. Each handles a distinct stage of the process. This modular design means any single stage can be updated, replaced, or scaled without touching the others.

WORKFLOW 01
SERAMAN Product Automation
The master orchestrator. Receives the Tally form submission, validates input, runs the Claude script agent, and calls the three sub-workflows in sequence.

WORKFLOW 02
Seraman Generate Videos
Dual-branch parallel generation engine. Runs Branch A (scenes 1 and 8) and Branch B (scenes 2–7) simultaneously, polls Kie AI for results, and writes all video URLs to the tracking sheet.

WORKFLOW 03
Seraman Edit Videos
Creatomate assembly pipeline. Reads all scene video URLs from the sheet, builds the Creatomate render payload, polls until assembly completes, and stores the final video URL.

WORKFLOW 04
SERAMAN Error Handler
Global error monitor. Fires automatically when any workflow in the n8n instance encounters an error. Sends a branded HTML email with the exact failure details and a direct link to the execution.

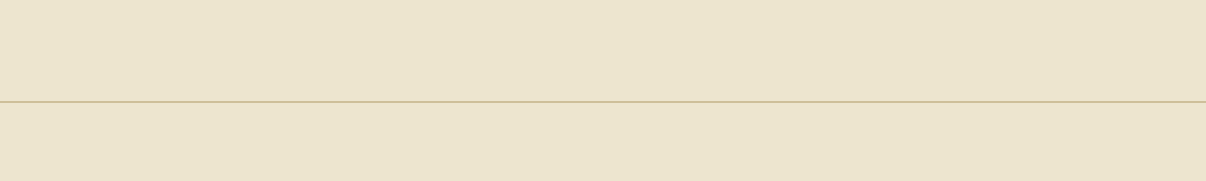


↑ SERAMAN Product Automation – master orchestration workflow

VIDEO GENERATION + ASSEMBLY

Generate, Assemble, Deliver

The Generate Videos workflow runs two branches in parallel — Branch A for the cinematic opener and branded close, Branch B for the six presenter-led scenes — each with its own async polling state machine. Once all eight scenes are ready, the Edit Videos workflow builds the Creatomate render payload, submits assembly, and polls every 60 seconds until the final MP4 is ready.



↑ Generate Videos – dual-branch Kie AI generation

↑ Edit Videos – Creatomate assembly pipeline

TRACKING AND LOGGING

Your Google Sheet

Every pipeline run is fully logged across three sheets. You can check the status of any product run at any time by opening the sheet.

Sheet 1
Generation tracking. Every scene's image prompt, video prompt, reference image URL, product image URL, and generation status. Updated in real time as each scene completes.

Sheet 2
Scene video URLs. The raw Kie AI output URL for each scene, keyed by scene number. Used by the Edit Videos workflow to build the Creatomate payload.

Sheet 3
Final video log. The assembled Creatomate video URL and the original Italian product description, one row per completed product run.

↑ Sheet 1 – scene-by-scene generation tracking with prompts and status

MONITORING

Automated Email Alerts

You receive a branded email notification after every pipeline run — whether it succeeds or fails. No need to check n8n manually.

WORKFLOW_SUCCEEDED
The pipeline ran cleanly. Your video is live. No action needed. The email confirms completion with no errors.

WORKFLOW_FAILED
Something stopped the pipeline. The email shows the exact workflow name, execution ID, the node that failed, and the error message. A direct link takes you straight to that execution in n8n.

↑ Success notification – WORKFLOW_SUCCEEDED

↑ Error notification – WORKFLOW_FAILED with execution link

ENGINEERING DEPTH

Under the Hood

What follows is a summary of the non-obvious engineering decisions built into this system — the problems that don't appear in demos but would surface in production at scale.

ASYNCHRONOUS STATE MACHINES WITH SELF-HEALING RETRY

Kie AI video generation is asynchronous — submitting a scene returns a task ID, not a video. The system runs an independent polling state machine per branch that checks every 30 seconds. Each scene tracks two counters: **retryCount** (up to 20 polling attempts — roughly 60 minutes of patience) and **regenCount** (up to 3 full scene regenerations if Kie AI reports a generation failure). A three-way router classifies every poll response into **success**, **failed**, or **pending** and routes each path differently. Failed scenes are flagged with a reason and written to the sheet — they never silently disappear.

ITEM IDENTITY INTEGRITY ACROSS ASYNC LOOPS

n8n's default item-linking is positional — it assumes item N going in maps to item N coming out. This breaks the moment a failed scene gets resummitted and takes a different path through the workflow at a different time. The fix: **scene_number travels forward as explicit data on every item** through every node. Results are always attributed to the correct scene regardless of what order they return from Kie AI. This eliminates an entire class of silent misattribution bugs that would cause wrong video clips to appear in the wrong scene slot.

CLAUDE V4 TRUST-FIRST SCRIPT ENGINEERING

The Claude script agent went through four major versions. Version 4 was rebuilt around one principle: **the presenter speaks like someone who has handled hundreds of products and has nothing to prove**. The prompt enforces a Problem → Experience → Feature → Benefit formula per scene, uses scene chaining so each scene connects naturally to the previous, applies buyer psychology ordering across the feature sequence, and runs an internal Trust Score Validator before output. A full banned language list hard-blocks words like 'premium', 'revolutionary', and 'mission-ready' — terms that trigger skepticism in tactical audiences.

- Strict JSON schema output — no markdown, no preamble, machine-parseable
- Product physics classification — boots, medical, optics, blades each handled differently
- Scene chaining — every scene opens with a connector to the previous
- Buyer psychology ordering — comfort, protection, performance, construction, certification

FAULT TOLERANCE – NO SCENE IS EVER LOST

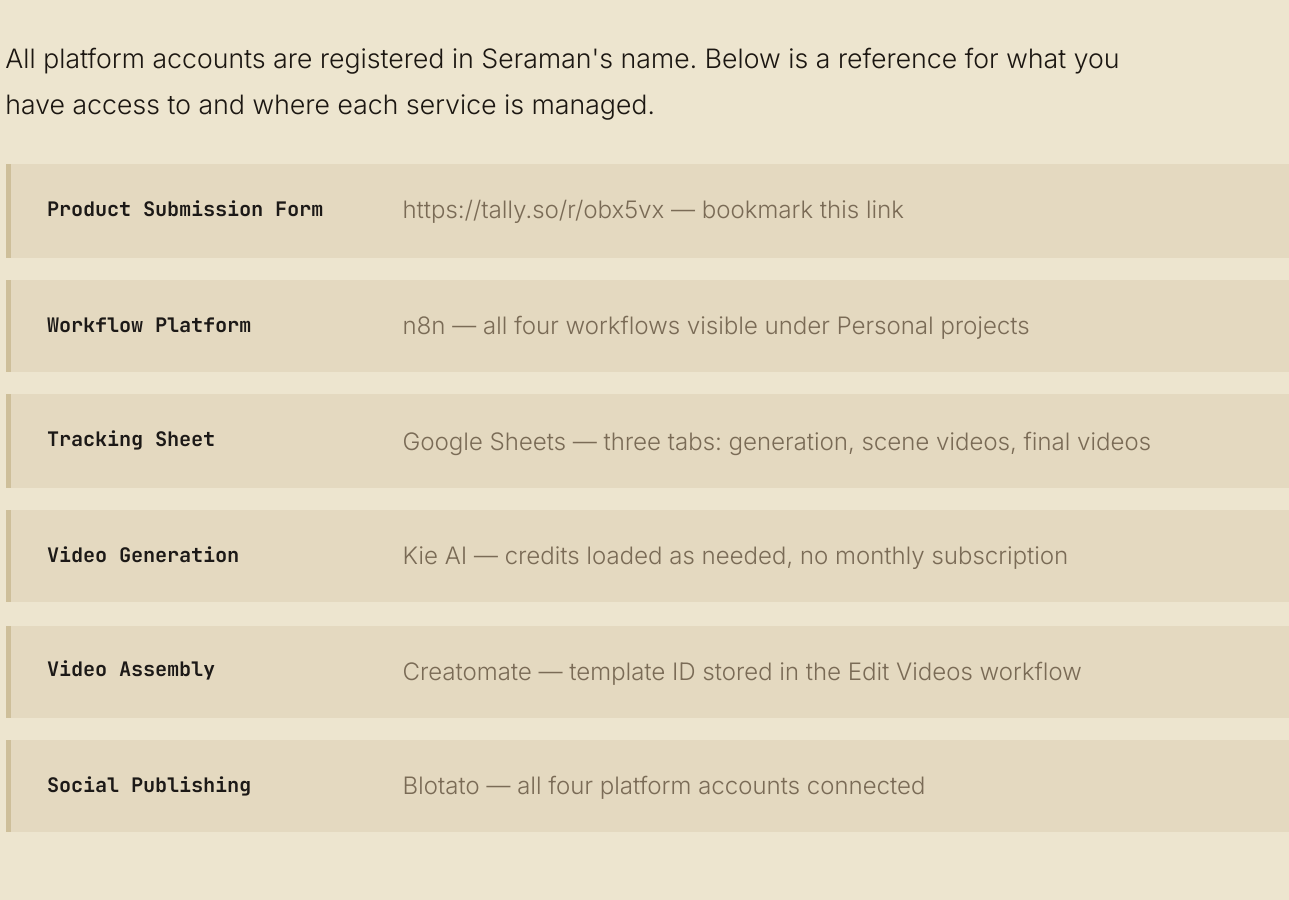
If one scene out of eight fails irrecoverably, the other seven still complete, get assembled, and get published. Both branches converge through a Merge node, get sorted back into correct scene order, and write to the output sheet as a complete set — success and failure rows alike. The entire batch never errors out because of a single scene failure. Failed scenes are clearly marked in the sheet for review.

ERROR HANDLING

Global Error Monitor

The Error Handler workflow runs independently of the main pipeline. It fires automatically whenever any workflow in the n8n instance throws an error — not just the video pipeline.

The email it sends is fully branded and built for debugging: JetBrains Mono font, dark matte design, highlighted error box, and a one-click link directly to the failed execution.



↑ SERAMAN Error Handler – global monitor for all workflow failures